

A self-organising Thucydides trap — v2

dynamic growth, deliberation by rollout, and war-driven alliances

M. Heltberg

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Self-contained. Supersedes the v1 model (see `NOTE_selforg.tex` for the earlier lattice-free version and its motivation). Code: `thucydides_selforg_v2.py`; figures `make_figures_v2.py`; depth sweep `experiments_v2.py`.

This version was built around four design decisions and one new mechanism that the v1 model lacked:

- growth rates that *vary in time* (so any state can rise, not only the large ones);
- no per-state capacity parameter;
- prediction by explicit *Monte-Carlo rollout* of future wars;
- **war-driven absorption**: the defeated side is swallowed into the victor’s alliance — which is the concrete reason allies fight (to avoid losing their partners to the enemy bloc).

The Q&A in §?? answers four specific questions; §?? gives the war rules in full and §?? a fully worked numerical example.

1 State and time

$N = 100$ states. State i carries

$S_i \geq 0$	strength
$\gamma_i(t)$	instantaneous growth rate (a slow stochastic process, below)
$x_i \in [0, 1)$	position on a 1-D ring (geography/ideology)
n_i	deliberation depth (number of rollouts it can run)
$b(i)$	alliance label; an <i>alliance</i> (bloc) is the set $\{j : b(j) = b(i)\}$
$\hat{S}_i$	its online estimate of its own growth slope

Time advances in *rounds* of length $\Delta = n_{\text{sub}} dt = 1$. Each round: (1) integrate the growth SDE for n_{sub} Euler steps; (2) update slope estimates; (3) alliance reconsideration; (4) war decisions; (5) predation; (6) elimination/respawn.

2 Growth: resource-bounded, with time-varying rates

$$dS_i = \left[\gamma_i(t) S_i \left(1 - \frac{S_i}{c_0} \right) R(t) - \delta S_i \right] dt + \sigma \sqrt{S_i} dW_i, \quad R(t) = \max\left(0, 1 - \frac{\sum_j S_j}{K}\right). \quad (1)$$

- **Time-varying rate.** $\gamma_i(t)$ is a slow Ornstein–Uhlenbeck process with a *common* mean, $d\gamma_i = \theta_g(\mu_g - \gamma_i)dt + \sigma_g dW_i^g$ (clipped to $[g_{\min}, g_{\max}]$, $\theta_g^{-1} \approx 80$ rounds). No state has a permanent edge: a state in a high- γ epoch rises, in a low- γ lull it stalls or shrinks. *Minor states can become great powers and vice versa*—rank mobility is built in (this is point 1 of the review).
- **No per-state capacity.** The heterogeneous c_i of v1 is gone. A single *common* saturation scale c_0 (one number, identical for every state—a “largest realistic size”, not a frozen hierarchy) is kept only to stop the shared-resource term from collapsing to a single winner (competitive exclusion). This is point 2: c_0 is a scale, not heterogeneity.
- **Shared resource.** $R(t)$ is the global resource ceiling: growth halts as total strength approaches K — no infinite growth.
- **Noise.** Demographic ($\sqrt{S}$, CIR-type) noise keeps $S \geq 0$.

3 Ring geometry

Each state sits at x_i on a ring; the arc distance is $d_{ij} = \min(|x_i - x_j|, 1 - |x_i - x_j|) \in [0, \frac{1}{2}]$ and proximity $\pi_{ij} = e^{-d_{ij}/\ell}$. Distance enters in two places: a far-away strong power is felt as less threatening (§??), and two blocs can only fight if their nearest members are within r_{war} (§??). Locality lets the world organise into *regional* blocs (a multipolar phase); a power strong enough to threaten across the ring overrides locality and forces global bipolarisation.

4 Predictability = deliberation by Monte-Carlo rollout

A state does not read a single forecast; it *simulates*. To evaluate any war it would start or join, a state of depth n_i runs n_i independent rollouts of the outcome and acts on the *average* utility. This is the heart of “predictability” in v2.

What one rollout contains. The decider knows its own strength S_i exactly but is *uncertain about others*. In rollout r it draws a noisy world,

$$S_{\text{own}}^{(r)} = S_{\text{own}}^{\text{other}} (1 + \varepsilon_1^{(r)}), \quad S_{\text{opp}}^{(r)} = S_{\text{opp}} (1 + \varepsilon_2^{(r)}), \quad \varepsilon^{(r)} \sim \mathcal{N}(0, \sigma_{\text{world}}^2), \quad (2)$$

where $S_{\text{own}}^{\text{other}}$ is the strength of its own side *excluding itself* and S_{opp} the opposing side. It then evaluates the two options (below). Because the world is noisy, a single rollout is a poor estimate; the mean over n_i rollouts has variance $\propto 1/n_i$, so **deeper deliberation yields a sharper expected-utility estimate and better decisions**. A strength tax $\propto n_i$ (an attention budget) is charged each round.

The utility: security = your coalition’s share of power. A state’s payoff is the *security* of its alliance, i.e. its coalition’s share of local power, valued at v_{sec} per unit of its own strength, minus the blood price of fighting and the autonomy cost of being absorbed. With Lanchester win probabilities (exponent q)

$$p_{\text{join}} = \frac{(S_{\text{own}} + S_i)^q}{(S_{\text{own}} + S_i)^q + S_{\text{opp}}^q}, \quad p_{\text{abst}} = \frac{S_{\text{own}}^q}{S_{\text{own}}^q + S_{\text{opp}}^q}, \quad (3)$$

and $V = v_{\text{sec}} S_i$, the per-rollout utilities are

$$U_{\text{join}} = V[p_{\text{join}}(1 - P_{\text{lo}}) + P_{\text{lo}}] - p_{\text{join}} \ell_w S_i - (1 - p_{\text{join}})(\ell_\ell + \beta) S_i, \quad (4)$$

$$U_{\text{abst}} = \begin{cases} V[p_{\text{abst}}(1 - P_{\text{lo}}) + P_{\text{lo}}] & (\text{ally: war happens without me}) \\ V \cdot (\text{projected power share}) & (\text{initiator: no war}) \end{cases} \quad (5)$$

Reading (??): *win* $\Rightarrow$ your coalition absorbs the enemy and you reach near-total local security $(1 - P_{\text{lo}})$, paying the small blood price ℓ_w ; *lose* $\Rightarrow$ you are absorbed (residual security P_{lo}) and pay ℓ_ℓ plus the autonomy cost β . The decision is

$$\boxed{\text{act (join / attack)} \iff \overline{U_{\text{join}}} > \overline{U_{\text{abst}}}} \quad (6)$$

(means over the n_i rollouts). The driver is $V(p_{\text{join}} - p_{\text{abst}})$: *pivotality* $\times$ security gain, weighed against the blood price. An ally joins precisely when its weight tips the balance *and* the alternative (its side losing and being absorbed) is bad — this is the “don’t lose your allies” incentive made quantitative.

5 Alliance formation (balancing on the ring)

Each round, every state reconsiders its bloc with probability ρ . A reconsidering state i forms *proximity-weighted* perceived bloc strengths $\tilde{\Sigma}_B = \sum_{j \in B} \pi_{ij} \hat{S}_j$ (a distant bloc counts less) and its top threat $T = \arg \max_{B \neq b(i)} \tilde{\Sigma}_B$. It then applies, in order:

- R1 Salience.** If $\tilde{\Sigma}_T < \theta \pi_{ii} \hat{S}_i$ (no nearby threat worth the trouble) $\Rightarrow$ stand alone. Diffuse power $\Rightarrow$ many states unthreatened $\Rightarrow$ multipolar; a risen hegemon is a salient threat to many $\Rightarrow$ they converge on one balancing coalition.
- R2 Dissolution.** Else if i ’s own bloc already dwarfs the threat ($\tilde{\Sigma}_{b(i)} > 1.8 \tilde{\Sigma}_T$) $\Rightarrow$ defect. This is what makes over-grown (post-conquest) coalitions *fragment*.
- R3 Balancing.** Else join the non-threat bloc whose strength, with i added, best matches the threat from above (aggregate just enough to deter T).
- R4 Bandwagon.** If the threat is overwhelming ($\tilde{\Sigma}_T > 2.5 \pi_{ii} \hat{S}_i$), with small probability p_{bw} *join* it instead (buy protection rather than resist).

6 War: trigger, joining, cost, absorption

Wars are *decisions*, not random events. Each round, for every pair of strongest blocs that are (i) spatially adjacent ($d < r_{\text{war}}$) and (ii) not hopelessly lopsided (strength ratio > 0.25), and with per-round opportunity p_{att} :

1. **Initiation (preventive).** The stronger bloc’s lead state i deliberates (§??, `as_initiator`): it attacks iff $\overline{U_{\text{join}}} > \overline{U_{\text{abst}}}$, where the no-war baseline in (§?) uses the *projected* power share. If the rival is rising, the projected share falls below what i can secure by winning *now* — so i strikes pre-emptively. This is the Thucydides trigger, internal to the rollout.
2. **Joining (the alliances are asked).** War starts between lead states i (side A) and j (side B). Every other member k of i ’s alliance decides *independently*, by its own rollout (§??),

whether to join side A ; likewise each member of j 's alliance for side B . Member k joins iff $\overline{U}_{\text{join}} > \overline{U}_{\text{abst}}$ with $S_{\text{own}}^{\text{other}} =$ (its side's strength so far) and $S_{\text{opp}} =$ (opposing lead's strength). It joins when it is *pivotal* and the prospect of its side losing—and being absorbed—is worse than the blood price.

3. **Resolution (Lanchester cost).** Let the final coalitions have strengths S_A, S_B . Side A wins with probability $S_A^q / (S_A^q + S_B^q)$. Then *both* sides pay strength:

$$\text{losers: } S_k \rightarrow (1 - \ell_\ell) S_k, \quad \text{winners: } S_k \rightarrow (1 - \ell_w) S_k, \quad \ell_\ell > \ell_w, \quad (7)$$

and the winners share spoils equal to a fraction ξ_{sp} of the strength destroyed on the losing side, distributed $\propto$ contribution.

4. **Absorption (the new rule).** The defeated belligerents—the loser lead and every ally that fought on the losing side—are *relabelled into the winner's alliance*: $b(k) \rightarrow b(\text{winner})$ for all losing participants. The vanquished become subordinate members of the victor's bloc. (Allies that stayed out keep their label; their now-leaderless bloc typically fragments next round via R1/R2.)

Absorption is what couples wars back to the alliance graph: winning *grows* your coalition, losing *dissolves* it into your enemy's. That is the strategic reason to fight at all—and the reason an ally joins rather than free-ride: if it lets its side lose, it loses its partners (and itself) to the enemy and faces a far stronger neighbour.

7 Worked example

Setup. Established power i ($S_i = 25$) leads bloc $A = \{i, a\}$ with ally a ($S_a = 15$), so $\Sigma_A = 40$. Rising challenger j ($S_j = 20$, growing fast) leads bloc $B = \{j, b\}$ with ally b ($S_b = 10$), $\Sigma_B = 30$. They are adjacent on the ring. Take $q = 1.6$, $v_{\text{sec}} = 4$, $\ell_w = 0.12$, $\ell_\ell = 0.45$, $\beta = 0.1$, $P_{\text{lo}} = 0.1$.

1. **i decides to attack.** Now, i 's share is $40/70 = 0.57$, but j is rising and i projects its share falling to ≈ 0.45 within the horizon. Win probability of A vs B : $40^{1.6} / (40^{1.6} + 30^{1.6}) = 365 / (365 + 231) = 0.61$. With $V = 4 \times 25 = 100$,

$$U_{\text{attack}} = 100[0.61(0.9) + 0.1] - 0.61(0.12)(25) - 0.39(0.55)(25) \approx 58, \quad U_{\text{no war}} = 100(0.45) = 45.$$

$57.7 > 45$: the *preventive* war is worth it (a static rival, share 0.57, would give $U_{\text{no war}} = 57$ and i would *not* attack — it is the rise that triggers war).

2. **Ally a is asked to join side A .** Its side without it is just i (25); the opposing lead is j (20). Joining raises the win probability

$$p_{\text{abst}} = \frac{25^{1.6}}{25^{1.6} + 20^{1.6}} = 0.59 \longrightarrow p_{\text{join}} = \frac{40^{1.6}}{40^{1.6} + 20^{1.6}} = 0.75.$$

a is *pivotal* (+0.16). With $V_a = 4 \times 15 = 60$, $U_{\text{join}} = 60[0.75(0.9) + 0.1] - 0.75(0.12)(15) - 0.25(0.55)(15) \approx 43.1$ vs $U_{\text{abst}} = 60[0.59(0.9) + 0.1] \approx 37.9$. a joins — not for spoils, but because if A loses, a is absorbed into B . (A *non-pivotal* ally, whose side wins anyway, would instead free-ride.)

3. **Resolution.** Final $S_A = 40$, $S_B = 30$; A wins with prob 0.61 — say it does. Strengths: losers $S_j \rightarrow 20(0.55) = 11$, $S_b \rightarrow 10(0.55) = 5.5$; winners $S_i \rightarrow 25(0.88) = 22$, $S_a \rightarrow 15(0.88) =$

13.2. Destroyed loser strength = $0.45 \times 30 = 13.5$; spoils $\xi_{\text{sp}} \times 13.5 = 6.75$ split $\propto$ contribution give $S_i \approx 26.2$, $S_a \approx 15.7$.

4. Absorption. j and b (the losing belligerents) are relabelled into A . The new alliance is $A = \{i, a, j, b\}$ with strengths $\{26.2, 15.7, 11, 5.5\}$ — i 's coalition has *grown from two members to four* and the challenger is now its subordinate. This is the Melian/Peloponnesian pattern: the defeated power becomes a subject of the victor.

8 Predation and selection (brief)

A weak, *unaligned* state next to a much stronger nearby bloc is conquered with hazard $\propto (1 - 0.85 n_i/n_{\text{hi}})$ (foresight is protective). Any state with $S < \epsilon \langle S \rangle$ is eliminated and respawned as the offspring of a strength-weighted survivor (inheriting n_i , mutated), at a fresh ring position with γ reset to the mean. Depth n_i turns out to be *near-neutral* under selection here (most states are aligned, rarely exposed), so for the scaling study below we treat n as a frozen control parameter rather than an evolved trait.

9 Questions answered

Q1: Why is there always ≥ 2 blocs — is a single bloc not allowed? A single bloc ($N_{\text{eff}} = 1$, a universal empire) *is* allowed and is reachable: absorption lets a victor grow by swallowing losers, and in conquest cascades N_{eff} dips toward 1 (we observe minima ≈ 1.6). It is not an *absorbing* state, for three reasons that keep pulling the system back to ≥ 2 : (a) **balancing** (R1/R3): a dominant power is a salient threat to everyone, who coalesce against it; (b) **dissolution** (R2): an over-grown bloc sheds members — empires fragment; (c) **deterrence** in the war utility: attacking the last balanced rival risks defeat and absorption, so it is usually declined. Also, war *requires* two blocs by definition, so a momentary empire simply stops fighting until it fragments. Hence one sees rise *and fall* of near-hegemonies rather than a frozen $N_{\text{eff}} = 1$.

Q2: How does a state decide from the Monte-Carlo predictions? It runs n_i rollouts (§??), each scoring the two options (§?)–(§?) on a noisy sampled world, and picks the option with the larger *average* utility: attack/join iff $\overline{U_{\text{join}}} > \overline{U_{\text{abst}}}$. Deeper $n_i \Rightarrow$ lower-variance average $\Rightarrow$ choices closer to the true expected utility.

Q3: What is the cost of war, and what happens to strengths and alliances? See §?? and the worked example §??. In short: both sides pay Lanchester strength fractions ($\ell_\ell = 0.45$ losers, $\ell_w = 0.12$ winners); the winner additionally takes spoils; and — crucially — the **defeated side is absorbed into the winner's alliance**. The loser j does join i 's alliance, exactly as you anticipated, and this is the reason allies join wars: not to lose their partners (and themselves) to the enemy bloc.

(Δ vs τ_{ema}). $\Delta = 1$ is the round length (the decision clock); $\tau_{\text{ema}} = 8$ is the slope-filter memory (how many rounds the trend estimate averages). Different objects.

10 Parameters

Growth: $K=1500$, $c_0=120$, $\delta=0.035$, $\sigma=0.16$; OU rate $\mu_g=0.11$, $\theta_g=0.012$, $\sigma_g=0.035$. Ring: $\ell=0.22$, $r_{\text{war}}=0.27$. Forecast: $\tau_{\text{ema}}=8$, $h=40$, $\sigma_{\text{world}}=0.40$. Alliances: $\rho=0.15$, $\theta=1.3$, $p_{\text{bw}}=0.10$.

War: $p_{\text{att}}=0.12$, $q=1.6$, $\ell_\ell=0.45$, $\ell_w=0.12$, $\xi_{\text{sp}}=0.5$, $\beta=0.1$, $v_{\text{sec}}=4$, $P_{\text{lo}}=0.1$. Runs: 4000 rounds, $N=100$.

11 Results: Options A and B

Option A — bipolarisation precedes the large wars. With dynamic growth and the ring, polarization N_{eff} genuinely moves: the world wanders between regional multipolar phases (Fig. ??c) and bipolar / near-hegemonic moments ($N_{\text{eff}} \rightarrow 1.6$, Fig. ??d). The largest (systemic) wars break out of the more bipolar configurations: across seeds the pre-war N_{eff} is lower for systemic than for small wars, and war severity rises as $N_{\text{eff}}^{\text{pre}} \rightarrow 2$ (Fig. ??e). The signal is weaker than in the absorption-free variant (the conquest cascades muddy it) but survives.

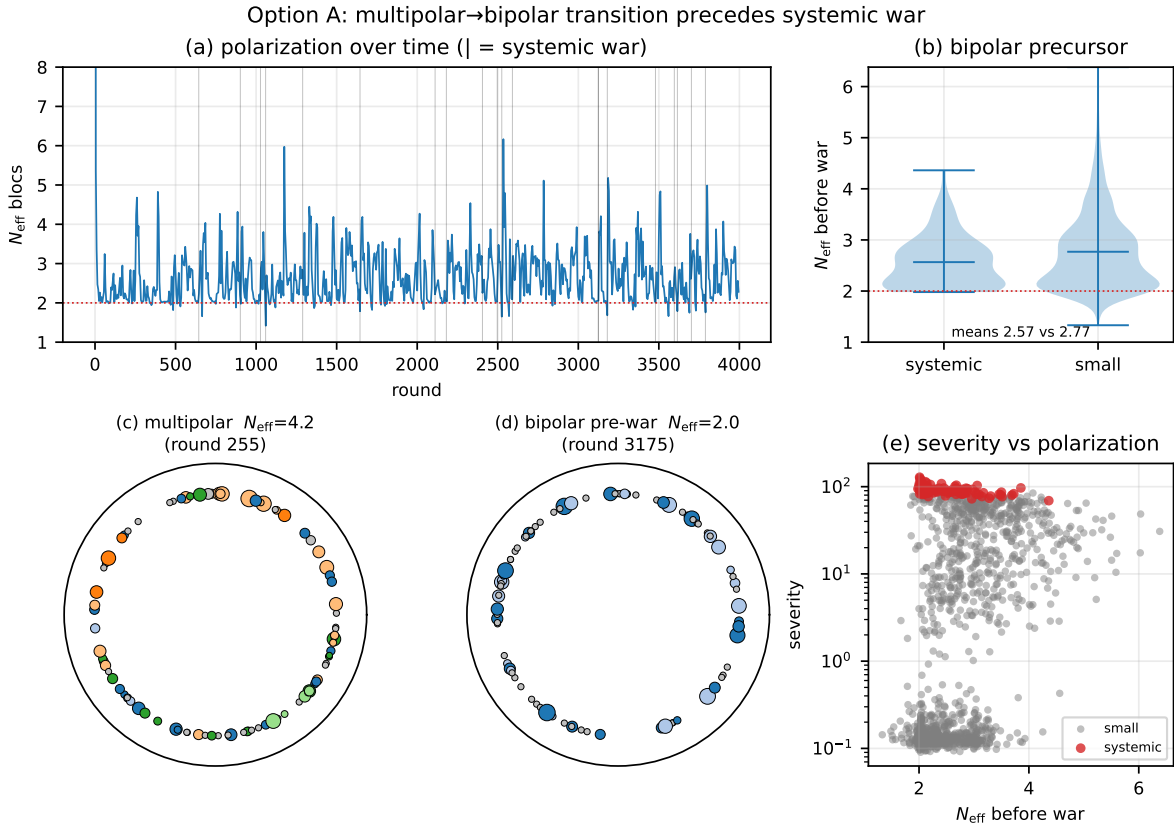


Figure 1: **Option A.** (a) $N_{\text{eff}}(t)$, 20 most severe wars marked; (b) pre-war N_{eff} , systemic vs small; (c) a multipolar phase and (d) a bipolar pre-war moment on the ring (dot size $\propto$ strength, colour = alliance); (e) severity vs pre-war N_{eff} .

Option B — with absorption, collective foresight is destabilising. Sweeping a frozen, homogeneous deliberation depth $n \in \{1, \dots, 64\}$ (3 seeds, 2500 rounds; Spearman ρ vs n):

- **Systemic-war rate** rises from 13 to 44 per 1000 rounds ($\rho = +0.96$).
- **Length of peace** τ falls from 66 to 22 rounds ($\rho = -0.96$).

- **Mean war severity** rises from 20 to 30 ($\rho = +0.93$).

This *inverts* the result of the absorption-free variant (where foresight kept the system multipolar and shrank wars). The reason is the new mechanism: when winning a war means *absorbing* your rival, a deeper-thinking state is a better predator — it more reliably identifies winnable conquests and the pivotal moments to join, and exploits them. Collective foresight therefore accelerates conquest and systemic war rather than deterring it.

With real conquest stakes, predictability is destabilising: the better states foresee the outcome of war, the more often the winnable war is fought, and the shorter the peace.

(Here polarization N_{eff} barely moves with n , $2.9 \rightarrow 3.1$; the lever is opportunism, not bloc structure.)

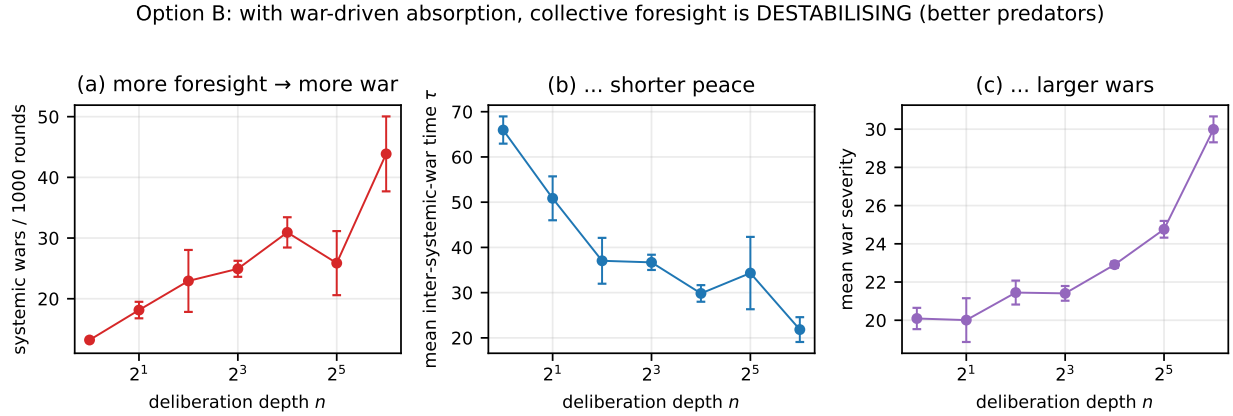


Figure 2: **Option B.** Deliberation depth n (log axis) as a control parameter; mean $\pm$ sd over seeds.

12 Outlook

The model now has the ingredients in the right places: endogenous rise-and-fall of powers (dynamic γ), prediction as simulation (rollouts), balancing alliances on a geography, and wars that reshape the alliance graph by absorption. Natural next steps: (i) a 2-bloc mean-field / first-passage theory for the inter-war time and the N_{eff} precursor; (ii) test whether war severity is a single function of $N_{\text{eff}}^{\text{pre}}$ (one-parameter law); (iii) strengthen the fitness coupling so deliberation depth self-organises to its cost-optimal value; (iv) confront NMC/COW data — is the major-power system near $N_{\text{eff}} \approx 2$ before 1914/1939, and is relative-capability *slope* the better war predictor?